# IMPLICIT DIFFERENTIATION

## DIFFERENTIAL CALCULUS

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# GRADING SYSTEM

# FINAL GRADE

30%

PRELIM

30%

MIDTERM

**40%**

ENDTERM

# TERM GRADE

60%

EXAM

40%

SEATWORK

# PENALTY

0

ABSENT

3

45-min LATE

5 max

15-min LATE

## TOPIC OUTLINE

# Implicit Differentiation

# IMPLICIT DIFFERENTIATION

# EXPLICIT FUNCTION

The dependent variable $(y)$ is expressed **directly** in terms of the independent variable $(x)$

Form

$$y = f(x)$$

$$y = 2x + 3$$

$$y = \sin x$$

# IMPLICIT FUNCTION

The relationship between the dependent variable $(y)$ and independent variable $(x)$ is not directly solved for one variable in terms of the other. Instead, the relationship is expressed as an **equation** involving **both variables**

Form

$$F(x, y) = 0$$

$$2x - y + 3 = 0$$

$$x^3 + y^3 = 6xy$$

If $x^2 + y^2 = 25$, find $\frac{dy}{dx}$.

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}(25)$$

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = 0$$

$$2x + 2y\frac{dy}{dx} = 0$$

$$2y\frac{dy}{dx} = -2x$$

$$\boxed{\frac{dy}{dx} = -\frac{x}{y}}$$

*ans*

$$F(x) = y^2$$

Let $u = y$ $\qquad f(u) = u^2$

$$F(x)' = \frac{df(u)}{du} \cdot \frac{du}{dx}$$

$$F(x)' = \frac{d}{du}u^2 \cdot \frac{du}{dx}$$

$$F(x)' = 2u \cdot \frac{du}{dx}$$

$$F(x)' = 2y \cdot \frac{dy}{dx}$$

Find $y'$ if $x^3 + y^3 = 6xy$

$$\frac{d}{dx}\left(x^3 + y^3\right) = \frac{d}{dx}(6xy)$$

$$\frac{d}{dx}\left(x^3\right) + \frac{d}{dx}\overset{u^3}{\left(y^3\right)} = y\frac{d}{dx}(6x) + 6x\frac{dy}{dx}$$

$$2x^2 + 3y^2\overset{3u^2\frac{du}{dx}}{\frac{dy}{dx}} = 6y + 6x\frac{dy}{dx}$$

$$3y^2\frac{dy}{dx} - 6x\frac{dy}{dx} = 6y - 2x^2$$

$$y^2\frac{dy}{dx} - 2x\frac{dy}{dx} = 2y - x^2$$

$$\frac{dy}{dx}\left(y^2 - 2x\right) = 2y - x^2$$

$$\boxed{\frac{dy}{dx} = \frac{2y - x^2}{y^2 - 2x}}$$

*ans*

Find $y'$ if $\cos x + \sqrt{y} = 5$

$$\frac{d}{dx}(\cos x + \sqrt{y}) = 5$$

$$\frac{d}{dx}(\cos x) + \frac{d}{dx}(\overset{u^{1/2}}{\sqrt{y}}) = \frac{d}{dx}(5)$$

$$-\sin x + \frac{1}{2}y^{-\frac{1}{2}}\overset{\frac{1}{2}u^{-1/2}\frac{du}{dx}}{\frac{dy}{dx}} = 0$$

$$\frac{1}{2\sqrt{y}}\frac{dy}{dx} = \sin x$$

$$\boxed{y' = 2\sqrt{y}\sin x}$$

*ans*

If $\cos(xy) = 1 + \sin y$, find $\frac{dy}{dx}$.

$$\frac{d}{dx}\cos(\overset{u}{xy}) = \frac{d}{dx}(1) + \frac{d}{dx}\sin y$$

$$-\sin u \frac{du}{dx}$$

$$-\sin(xy)\frac{d}{dx}(xy) = \cos y \frac{dy}{dx}$$

$$-\sin(xy)\left[y\frac{d}{dx}x + x\frac{d}{dx}y\right] = \cos y \frac{dy}{dx}$$

$$-\sin(xy)\left[y + x\frac{dy}{dx}\right] = \cos y \frac{dy}{dx}$$

$$-y\sin(xy) - x\sin(xy)\frac{dy}{dx} = \cos y \frac{dy}{dx}$$

$$-y\sin(xy) = \cos y \frac{dy}{dx} + x\sin(xy)\frac{dy}{dx}$$

$$-y\sin(xy) = \frac{dy}{dx}(\cos y + x\sin(xy))$$

$$\boxed{\frac{dy}{dx} = -\frac{y\sin(xy)}{x\sin(xy) + \cos y}}$$

*ans*

Use implicit differentiation to find the **equation** of the **tangent line** to the curve

$$x^2 + 2xy - y^2 + x = 2 \ @(1,2)$$

$$\frac{d}{dx}x^2 + \frac{d}{dx}(2xy) - \frac{d}{dx}y^2 + \frac{d}{dx}x = \frac{d}{dx}(2)$$

$$2x + y\frac{d}{dx}(2x) + 2x\frac{dy}{dx} - 2y\frac{dy}{dx} + 1 = 0$$

$$2x + 2y + 2x\frac{dy}{dx} - 2y\frac{dy}{dx} + 1 = 0$$

$$\frac{dy}{dx}(2x - 2y) = -2x - 2y - 1$$

$$\frac{dy}{dx} = -\frac{2x + 2y + 1}{2x - 2y}$$

$\underline{@(1,2)}$

$$\frac{dy}{dx} = -\frac{2(1) + 2(2) + 1}{2(1) - 2(2)}$$

$$\frac{dy}{dx} = -\frac{2 + 4 + 1}{2 - 4}$$

$$\frac{dy}{dx} = \frac{7}{2} \longleftarrow \text{slope}$$

Use implicit differentiation to find the **equation** of the **tangent line** to the curve

$$x^2 + 2xy - y^2 + x = 2 \ @(1, 2)$$

Point-slope form

$$y - y_1 = m(x - x_1)$$

$$y - 2 = \frac{7}{2}(x - 1)$$

$$2y - 4 = 7x - 7$$

$$2y = 7x - 3$$

$$\boxed{y = \frac{7}{2}x - \frac{3}{2}}$$

*ans*

# SEATWORK